A Geometric Hypothesis Connecting Black Holes and White Holes Through Ray-Like Spacetime Evolution

By Labonnozakiya

Abstract

Black holes are among the most fascinating predictions of Einstein's General Theory of Relativity, while white holes remain hypothetical objects with no confirmed observational evidence. In this article, I present a personal geometric hypothesis proposing that black holes and white holes may be interpreted as two stages of a continuous spacetime evolution connected through a singularity. Inspired by geometric reasoning and numerical visualization, this model introduces the idea that spacetime behaves like a propagating ray passing through a transition region. This work is exploratory and intended to encourage discussion rather than claim an established physical theory.

Introduction

Modern physics describes a black hole as a region where gravity becomes so strong that nothing, not even light, can escape after crossing the event horizon. A white hole is mathematically related to black holes but has never been observed.

This article explores a different question:

Can black holes and white holes represent different geometric phases of a single spacetime process?

Instead of treating them as completely separate objects, this hypothesis considers them parts of one continuous spacetime evolution.

The Ray Hypothesis

The central idea is simple.

Imagine spacetime behaving similarly to a ray.

As matter collapses under gravity, spacetime bends toward a singularity.

Instead of ending there, the singularity may represent a transition geometry that connects to an expanding spacetime region.

The proposed sequence is

Normal Spacetime

↓

Event Horizon

↓

Singularity

↓

White Hole Region

↓

Expanding Spacetime

This interpretation views singularity as a geometric bridge rather than only a point of infinite density.

Geometric Model

The hypothesis introduces three geometric parameters.

a = Event Horizon Radius

b = Distance toward Singularity

c = Expansion Geometry after Transition

The exploratory relationships suggested during this work include

$$a + b + c = n\sqrt{3}a$$

and

$$c - b = n\sqrt{3}$$

where **n** represents a geometric scaling parameter.

These expressions are currently conceptual and require rigorous mathematical derivation before being considered physical equations.

Ray-like Spacetime Evolution

Instead of imagining spacetime as static, this hypothesis proposes that it evolves continuously like the propagation of a wave or ray.

Black Hole

↓

Compression

↓

Transition at Singularity

↓

White Hole

↓

Expansion

Such a process could provide a new geometric interpretation of spacetime evolution.

Numerical Visualization

Python-based simulations were created to visualize spacetime curvature around a massive object.

The generated three-dimensional curvature illustrates how gravitational influence increases toward the center.

Although simplified, these simulations provide qualitative support for the geometric intuition behind this hypothesis.

Discussion

This work does **not** replace Einstein's General Relativity.

Instead, it attempts to explore whether an alternative geometric description could help visualize black hole–white hole transitions.

Several important questions remain open.

- Can Einstein Field Equations produce these relationships?
- Can the hypothesis conserve energy and momentum?
- Does quantum gravity support this interpretation?
- Can astronomical observations test this model?

Future mathematical work is required before these questions can be answered.

Future Work

The next stage of this research will include

- Tensor-based mathematical formulation
- Numerical simulations
- Comparison with Schwarzschild Geometry
- Analysis of spacetime curvature
- Investigation of possible observational predictions

Conclusion

Black holes and white holes remain among the greatest mysteries in theoretical physics.

This article presents a speculative geometric hypothesis suggesting that spacetime may evolve continuously through a ray-like transition connecting gravitational collapse with expansion.

Whether the hypothesis is ultimately correct or incorrect, exploring new mathematical perspectives can contribute to future discussions in theoretical physics.

Disclaimer

This article represents the author's personal theoretical hypothesis.

It has **not been peer-reviewed, experimentally verified, or accepted by the scientific community.**

Its purpose is to encourage scientific discussion and future mathematical investigation.